Transmission Mechanisms of Monetary Policy

Chapter 25

Econ 351 – Monetary Economics

University of Tennessee, Knoxville

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Preview

This chapter examines the **transmission mechanisms** of monetary policy — the channels through which central bank actions affect the real economy.

Central questions we will address:

1. How does a change in interest rates translate into changes in **output and employment**?
2. What role do **asset prices** (stocks, exchange rates, housing) play in transmitting monetary policy?
3. How do **credit market frictions** amplify or dampen policy effects?
4. What are the key **lessons** for conducting monetary policy?

Outline

How Does Monetary Policy Affect the Economy?

The Core Question

A **transmission mechanism** describes the chain of cause and effect from a **policy action** (e.g., the Fed lowers the federal funds rate) to a **change in aggregate demand and output**.

The Traditional Keynesian Channel

The simplest and most widely taught mechanism:

$$M\uparrow \Rightarrow\ i_r\downarrow \Rightarrow\ I\uparrow \Rightarrow\ Y_{ad}\uparrow$$

- $M\uparrow$: expansionary monetary policy
- $i_r\downarrow$: real interest rate falls
- $I\uparrow$: investment spending rises (business fixed investment, residential housing, consumer durables)
- $Y_{ad}\uparrow$: aggregate demand and output increase

Why Real Rates Matter

Real vs. Nominal Interest Rates

Spending decisions depend on the **real interest rate**, not the nominal rate:

$$r = i - \pi^e$$

Even when nominal rates are low, if π^e falls (or turns negative), the real rate can be **high** — tightening financial conditions.

Example

Japan in the 1990s had near-zero nominal rates, but deflation meant real rates were **positive and high**, keeping policy effectively tight.

With $i = 0$ and $\pi^e = -2\%$, the real rate is $r = 0 - (-2\%) = +2\%$: policy is **tight** even though the nominal rate is at its floor. What matters for spending is r, not i.

Why Long-Term Rates Matter

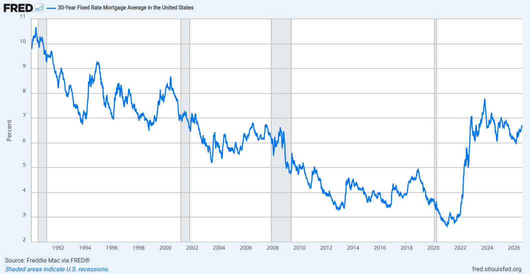
Short-Term vs. Long-Term Rates

- The Fed directly controls the **short-term** federal funds rate
- But investment and housing decisions depend on **long-term** real rates (e.g., 10-year Treasury, 30-year mortgage)
- The link between short and long rates depends on **expectations** of future policy (recall the expectations hypothesis from Chapter 24)

Implication

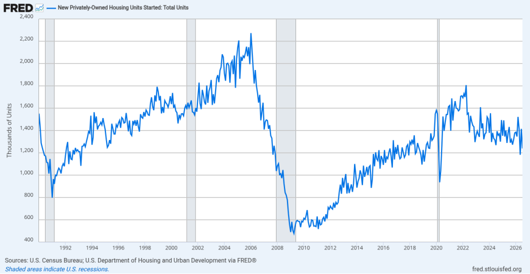
A cut in the federal funds rate will only stimulate spending if it also brings down **long-term real interest rates**. If markets expect the cut to be temporary, long rates may barely move.

The Interest-Rate Channel in the Data: Mortgages and Housing



30-year fixed mortgage rate, 1990–2025: the long-term rate that matters for housing decisions.

Source: FRED, Federal Reserve Bank of St. Louis



New housing starts (thousands of units), 1990–2025: residential investment collapses when borrowing costs rise and credit tightens.

Source: FRED, Federal Reserve Bank of St. Louis

Residential housing is one of the most interest-sensitive components of spending: a Fed cut stimulates it only if it also pulls down **long-term** rates such as the mortgage rate.

The Interest-Rate Channel: Easing vs. Tightening

Expansionary Policy

1. Fed lowers federal funds rate
2. Real interest rates fall
3. Cost of borrowing decreases
4. Business investment $\uparrow$
5. Residential housing $\uparrow$
6. Consumer durable spending $\uparrow$
7. **Aggregate demand** $\uparrow$

Contractionary Policy

1. Fed raises federal funds rate
2. Real interest rates rise
3. Cost of borrowing increases
4. Business investment $\downarrow$
5. Residential housing $\downarrow$
6. Consumer durable spending $\downarrow$
7. **Aggregate demand** $\downarrow$

Outline

Beyond Interest Rates: Multiple Channels

Key Insight

The interest-rate channel is **not the only** way monetary policy affects the economy. Changes in monetary policy also move **other asset prices**, each of which creates its own transmission channel.

Exchange Rate Channel

- Affects net exports
- Important for open economies

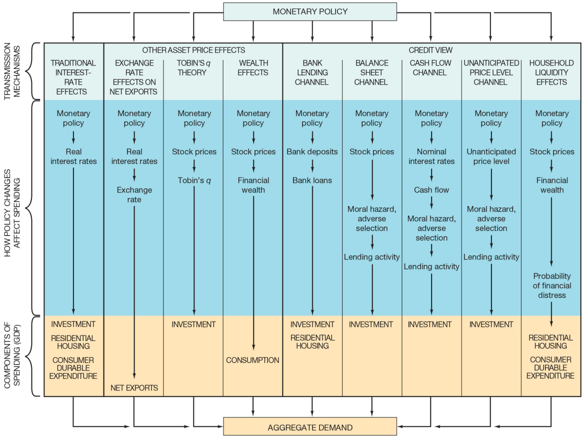
Tobin's q Channel

- Affects firm investment via stock prices

Wealth Effects Channel

- Affects consumption via household wealth

Monetary Transmission Mechanisms: The Big Picture



All major transmission channels from monetary policy to aggregate demand, including traditional interest-rate effects, other asset price effects, and credit view channels.

Mishkin, The Economics of Money, Banking, and Financial Markets

The Exchange Rate Channel

Mechanism

$$M\uparrow \Rightarrow \; i_r\downarrow \Rightarrow \; E\downarrow \; \text{(depreciation)} \; \Rightarrow \; NX\uparrow \Rightarrow \; Y_{ad}\uparrow$$

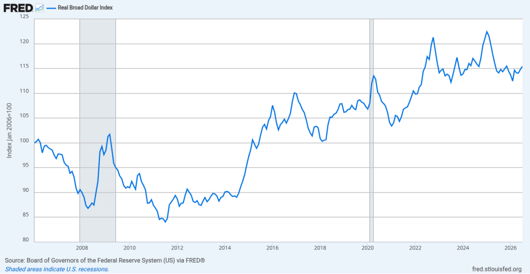
How It Works

1. Expansionary policy lowers domestic **real interest rates**
2. Domestic assets become **less attractive** relative to foreign assets
3. Capital flows out $\Rightarrow$ domestic currency **depreciates**
4. Domestic goods become **cheaper** for foreigners
5. **Net exports rise**, boosting aggregate demand

Importance

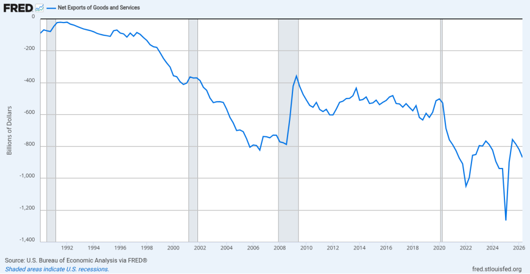
This channel is especially powerful for **small open economies** (e.g., Canada, Sweden, emerging markets) where trade is a large share of GDP.

The Exchange Rate Channel in the Data



Real broad dollar index, 2006–2025: a higher index means U.S. goods are dearer relative to foreign goods.

Source: FRED, Federal Reserve Bank of St. Louis



U.S. net exports of goods and services (billions of dollars), 1990–2025.

Source: FRED, Federal Reserve Bank of St. Louis

Lower domestic real rates $\Rightarrow$ capital outflow $\Rightarrow$ depreciation ($E \downarrow$) $\Rightarrow$ $NX \uparrow$. For the U.S., a large and relatively closed economy, this channel is weaker than for small open economies.

Tobin's q Theory: Definition

Definition (James Tobin)

$$q = \frac{\text{Market Value of Firms}}{\text{Replacement Cost of Capital}}$$

When $q > 1$

- Firms are valued **above** the cost of building new capital
- Issuing new equity is **cheap** relative to the capital it buys
- Firms **increase investment**

When $q < 1$

- Firms are valued **below** replacement cost
- Buying existing firms is **cheaper** than building new capital
- Firms **reduce investment**

Tobin's q: The Monetary Policy Link

Transmission Mechanism

$$M \uparrow \Rightarrow P_{\text{stocks}} \uparrow \Rightarrow q \uparrow \Rightarrow I \uparrow \Rightarrow Y_{ad} \uparrow$$

How It Works

1. Expansionary policy lowers interest rates
2. Bonds become less attractive $\Rightarrow$ investors shift to **stocks**
3. **Stock prices rise**
4. q increases $\Rightarrow$ new capital is cheap relative to market value
5. Firms **increase investment spending**

Key Insight

This channel works even if firms don't borrow at all — it operates through **equity markets** rather than debt markets.

The Wealth Effects Channel: Theory

Theoretical Foundation: Life-Cycle Hypothesis (Modigliani)

Household consumption depends on **lifetime resources**, not just current income. Financial wealth is a major component of lifetime resources.

Components of Financial Wealth

- Stocks and mutual funds
- Bonds and savings
- Real estate (housing)

Why Housing Matters

For most households, their home is their **largest asset**. Rising house prices increase perceived wealth and the ability to borrow against home equity.

The Wealth Effects Channel: Mechanism

Transmission Mechanism

$$M \uparrow \Rightarrow \; P_{\text{stocks}} \uparrow, \; P_{\text{housing}} \uparrow \Rightarrow \; \text{Wealth} \uparrow \Rightarrow \; C \uparrow \Rightarrow \; Y_{ad} \uparrow$$

Expansionary Policy

- Interest rates fall
- Stock and housing prices rise
- Household wealth increases
- **Consumption increases**

Contractionary Policy

- Interest rates rise
- Asset prices fall
- Wealth declines
- **Consumption decreases**

Outline

Why the Credit View Matters

Limitation of the Traditional View

The traditional interest-rate channel assumes that **all borrowers have equal access** to credit markets. In reality, **financial frictions** — adverse selection and moral hazard — create additional channels.

The Credit View

Financial frictions in credit markets create transmission channels that **amplify and propagate** the effects of monetary policy beyond what interest rates alone would predict.

Two Main Credit Channels

1. **Bank Lending Channel** — operates through the *supply* of bank loans
2. **Balance Sheet Channel** — operates through borrowers' *net worth*

The Bank Lending Channel

Mechanism

$$M\uparrow\Rightarrow\ \text{Bank Deposits}\uparrow\Rightarrow\ \text{Bank Loans}\uparrow\Rightarrow\ I\uparrow,\ C\uparrow\Rightarrow\ Y_{ad}\uparrow$$

How It Works

1. Expansionary policy increases **bank reserves**
2. Banks have more capacity to **make loans**
3. Borrowers who depend on bank credit (especially **small firms** and **households**) gain access to financing
4. Investment and consumption rise

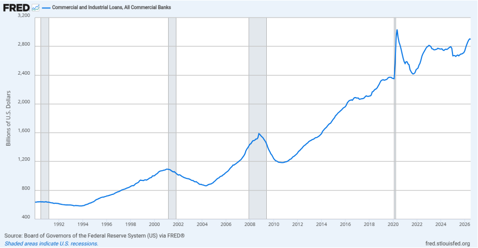
The Bank Lending Channel: Why It Is Distinct

Key Point

Even if market interest rates don't change much, an increase in the **quantity of bank lending** can stimulate spending — particularly for borrowers who **cannot access bond or equity markets** directly.

Who Is Most Affected?

- **Small and medium firms** that rely on bank loans as their primary funding source
- **Households** seeking mortgages, auto loans, and credit cards
- Borrowers in **developing economies** where capital markets are underdeveloped



Commercial and industrial loans at U.S. banks (billions of dollars), 1990–2025: bank credit contracts in each recession, most sharply after 2008.

Source: FRED, Federal Reserve Bank of St. Louis

The Balance Sheet Channel

Borrowers need strong **balance sheets** (high net worth, good collateral) to obtain loans. Monetary policy affects balance sheets through asset prices.

Mechanism

$$M \uparrow \Rightarrow P_{\text{assets}} \uparrow \Rightarrow \text{Net Worth} \uparrow \Rightarrow \text{Adverse Selection} \downarrow, \text{ Moral Hazard} \downarrow$$

$$\Rightarrow \text{Lending} \uparrow \Rightarrow I \uparrow \Rightarrow Y_{ad} \uparrow$$

Why Net Worth Matters

- Higher net worth means borrowers have more **"skin in the game"** $\Rightarrow$ less moral hazard
- Higher collateral values reduce **adverse selection** $\Rightarrow$ lenders are more willing to extend credit
- Result: credit flows more freely, amplifying the policy effect

The Cash Flow Channel

Mechanism

$$M\uparrow \Rightarrow \; i\downarrow \Rightarrow \; \text{Cash Flow}\uparrow \Rightarrow \; \text{Lending}\uparrow \Rightarrow \; I\uparrow$$

How It Works

- Lower nominal interest rates reduce firms' **debt service payments**
- Higher cash flow improves the firm's **financial position**
- Reduces adverse selection and moral hazard $\Rightarrow$ more lending

Example

A firm with $1 million in variable-rate debt: if rates fall by 2%, it saves $20,000/year in interest payments, improving its balance sheet and making lenders more willing to extend new credit.

Other Credit Channels

Unanticipated Price-Level Channel

- Unexpected inflation **reduces the real value of debt**
- Borrowers' real net worth **increases**
- Lending conditions improve $\Rightarrow$ investment rises

Household Liquidity Effects

- Higher asset prices increase household **liquid wealth**
- Households feel more financially secure $\Rightarrow$ spend more on **durables and housing**
- Reduced probability of **financial distress**

Why Credit Channels Are Important

Credit Channels Amplify Monetary Policy

Financial frictions create a **financial accelerator**: small changes in monetary policy can produce **large changes** in credit conditions, which amplify the effect on output.

Disproportionate Impact on Small Firms

- Small firms depend heavily on **bank loans** (no access to bond or equity markets)
- They have **fewer financing alternatives** when credit tightens
- Monetary policy therefore has a **larger effect** on small firms than on large corporations

Empirical Evidence for the Credit View

Research Findings

Studies show that after a monetary tightening:

- Small firms cut investment **more** than large firms
- Bank-dependent borrowers are affected **first and most severely**
- These patterns are consistent with the credit view

Implication

The credit view helps explain why monetary policy can have **larger real effects** than the traditional interest-rate channel alone would predict — especially during financial crises when frictions are most severe.

The Role of Financial Frictions

Information Problems in Credit Markets

- **Adverse selection**: before the loan is made, lenders cannot perfectly distinguish good from bad borrowers
- **Moral hazard**: after the loan is made, borrowers may take excessive risks with the lender's money

How Banks Mitigate These Problems

Banks are specialists in overcoming information asymmetries:

- **Screening** borrowers before lending
- **Monitoring** borrowers during the loan
- Requiring **collateral** and restrictive covenants
- Building long-term **relationships** with borrowers

When the Banking System Breaks Down

Implication for Monetary Policy

When monetary policy disrupts the banking system (e.g., a credit crunch), these information-processing functions break down, and the real economy suffers **disproportionately**.

Example: 2008 Financial Crisis

- Banks suffered massive losses on mortgage-backed securities
- They **tightened lending standards** dramatically
- Even creditworthy borrowers were **denied loans**
- The credit channel breakdown turned a financial crisis into a **deep recession**

Outline

All Channels at a Glance

Channel	Key Link	Affects
Interest-rate	$i_r \downarrow \Rightarrow I \uparrow$	Investment, durables, housing
Exchange rate	$E \downarrow \Rightarrow NX \uparrow$	Net exports
Tobin's q	$P_{\text{stocks}} \uparrow \Rightarrow q \uparrow \Rightarrow I \uparrow$	Business investment
Wealth effects	Wealth $\uparrow \Rightarrow C \uparrow$	Consumption
Bank lending	Loans $\uparrow \Rightarrow I, C \uparrow$	Investment, consumption
Balance sheet	Net worth $\uparrow \Rightarrow$ Lending $\uparrow$	Investment
Cash flow	Cash flow $\uparrow \Rightarrow$ Lending $\uparrow$	Investment
Household liquidity	Liquid assets $\uparrow \Rightarrow C \uparrow$	Durables, housing

Key Takeaway

Monetary policy works through **many channels simultaneously**. No single channel tells the whole story — the combined effect is what matters for aggregate demand.

Outline

The Traditional Interest-Rate Channel

Other Asset Price Channels

The Credit View

Summary of All Transmission Channels

Lessons for Monetary Policy

Lesson 1: Nominal Rates Are a Poor Policy Indicator

The Common Mistake

Judging the stance of monetary policy by looking at **short-term nominal interest rates** alone.

Why This Is Misleading

- The relevant variable for spending decisions is the **real interest rate**: $r = i - \pi^e$
- If inflation expectations fall (or deflation sets in), real rates can be **high even when nominal rates are near zero**
- Other asset prices (exchange rates, stock prices, credit spreads) may tell a **different story** about financial conditions

Lesson 2: Watch Other Asset Prices Too

Beyond the Federal Funds Rate

Because monetary policy operates through **multiple channels**, policymakers should monitor a **broad set of asset prices**:

Stock Prices

- Signal Tobin's q
- Reflect wealth effects
- Indicate business confidence

Exchange Rates

- Signal competitiveness
- Affect net exports
- Reflect capital flows

Housing Prices

- Affect household wealth
- Influence collateral values
- Drive residential investment

$\Rightarrow$ No single indicator captures the full stance of monetary policy. Policymakers need a **dashboard**, not a single gauge.

Lesson 3: Policy Still Works at the Zero Lower Bound

The Concern

When nominal interest rates hit zero, the traditional interest-rate channel appears to be **exhausted**. Does monetary policy become powerless?

The Answer: No

The existence of **multiple transmission channels** means policy can still work through:

- **Quantitative easing (QE)**: purchasing long-term assets to lower long-term rates and credit spreads
- **Forward guidance**: committing to keep rates low to influence expectations and long-term rates
- **Credit channel effects**: QE increases bank reserves, supporting lending

Lesson 3 (continued): More Channels at the ZLB

More Channels That Work at the ZLB

- **Exchange rate effects**: QE can weaken the currency, boosting net exports
- **Wealth effects**: asset purchases raise stock and bond prices, increasing household wealth and consumption
- **Balance sheet effects**: rising asset prices improve borrowers' net worth, easing credit conditions

Example

The Federal Reserve used QE and forward guidance in 2009 and again during the COVID-19 crisis in 2020, operating through all of these non-interest-rate channels simultaneously.

Lesson 4: Avoid Unanticipated Price-Level Fluctuations

Why Price Stability Matters for Transmission

Unanticipated changes in the price level disrupt the credit channels:

Unexpected Deflation

- Increases the **real burden of debt**
- Reduces borrowers' **net worth**
- Worsens adverse selection and moral hazard
- Credit contracts $\Rightarrow$ **debt deflation spiral**

Unexpected Inflation

- Increases **economic uncertainty**
- Distorts price signals
- Makes long-term planning **difficult**
- Can lead to **financial instability**

Lesson

Stable, predictable inflation keeps the credit channels functioning smoothly and prevents the **financial accelerator** from amplifying shocks.

Application: The 2008 Financial Crisis

What Happened

The collapse of the U.S. housing market triggered a **breakdown in multiple transmission channels simultaneously**:

Channel Disruptions

- **Balance sheet channel**: falling house prices destroyed household and firm net worth
- **Bank lending channel**: banks suffered massive losses and **cut lending sharply**
- **Wealth effects**: stock market crash reduced household wealth by trillions
- **Cash flow channel**: rising defaults increased financial frictions

Application: The 2008 Financial Crisis (continued)

Policy Response

The Fed used **every available tool**:

- Rate cuts to **zero**
- **QE**: over $4 trillion in asset purchases
- **Emergency lending facilities** for banks and financial institutions
- **Forward guidance**: committing to keep rates low for an extended period

Why So Many Tools?

Because **multiple channels** had broken down simultaneously, no single tool was sufficient. The Fed needed to address the interest-rate channel, credit channels, and wealth effects **all at once**.

Chapter Summary

1. The **traditional interest-rate channel** works through real long-term interest rates affecting investment, housing, and durable goods spending.
2. **Other asset price channels** — exchange rates, Tobin's q, and wealth effects — provide additional pathways from monetary policy to aggregate demand.
3. The **credit view** (bank lending channel, balance sheet channel, cash flow channel, household liquidity effects) shows that financial frictions **amplify** the effects of monetary policy.
4. **Lesson 1**: Nominal interest rates alone are a **poor indicator** of the stance of monetary policy.
5. **Lesson 2**: Policymakers should monitor a **broad set of asset prices**.
6. **Lesson 3**: Policy remains effective at the **zero lower bound** through unconventional tools.
7. **Lesson 4**: **Price stability** is essential to keep credit channels functioning.

Key Transmission Equations (1)

Traditional Interest-Rate Channel

$$M\uparrow \Rightarrow\ i_r\downarrow \Rightarrow\ I\uparrow \Rightarrow\ Y_{ad}\uparrow$$

Exchange Rate Channel

$$M\uparrow \Rightarrow\ i_r\downarrow \Rightarrow\ E\downarrow \Rightarrow\ NX\uparrow \Rightarrow\ Y_{ad}\uparrow$$

Tobin's q Channel

$$M\uparrow \Rightarrow\ P_{\text{stocks}}\uparrow \Rightarrow\ q\uparrow \Rightarrow\ I\uparrow \Rightarrow\ Y_{ad}\uparrow$$

Key Transmission Equations (2)

Wealth Effects Channel

$$M \uparrow \Rightarrow \; P_{\text{assets}} \uparrow \Rightarrow \; \text{Wealth} \uparrow \Rightarrow \; C \uparrow \Rightarrow \; Y_{ad} \uparrow$$

Bank Lending Channel

$$M \uparrow \Rightarrow \; \text{Deposits} \uparrow \Rightarrow \; \text{Loans} \uparrow \Rightarrow \; I, C \uparrow \Rightarrow \; Y_{ad} \uparrow$$

Balance Sheet Channel

$$M \uparrow \Rightarrow \; P_{\text{assets}} \uparrow \Rightarrow \; \text{Net Worth} \uparrow \Rightarrow \; \text{Lending} \uparrow \Rightarrow \; I \uparrow \Rightarrow \; Y_{ad} \uparrow$$